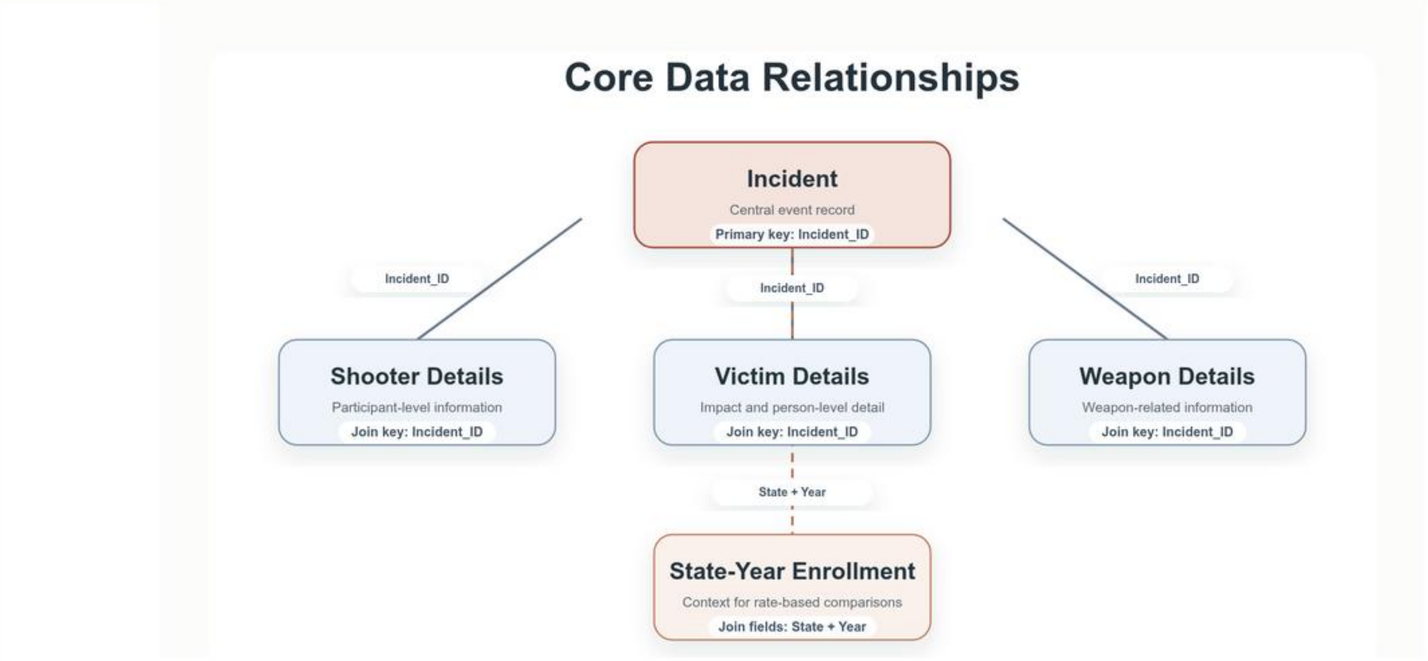
Data Structure

Core Data Relationships

The project combines event-level detail with state-year context. Incidents anchor the relational tables, while enrollment and policy joins create the panel used for risk analysis.



Structure of the data

- The incident table is the central event record for the project.
- Shooter, victim, and weapon records add incident-level detail through Incident_ID.
- Enrollment and policy tables attach state-year context for normalized comparisons.

3,136
Incident rows in the anchor table.

2,548
Enrollment state-year rows used for exposure.

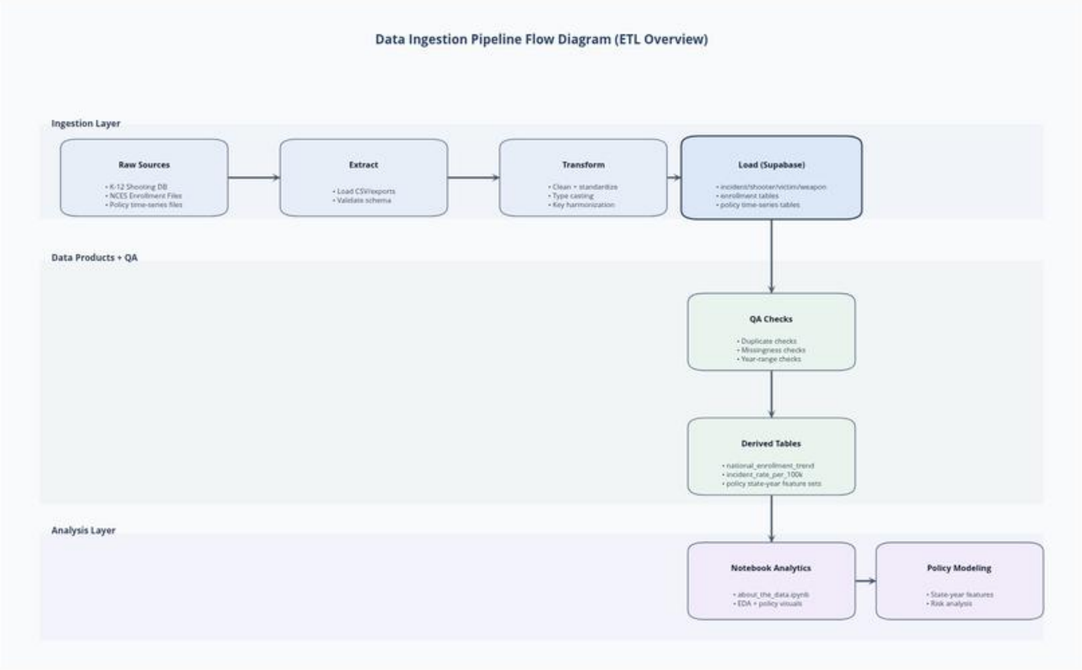
Analytic consequence
State-year joins turn event records into a normalized risk panel.

Key rate used throughout the project: incident_count / total_students x 100,000

Data Credibility

Analytic Pipeline And Quality Controls

The repo does substantial cleaning before modeling. It standardizes keys, rebuilds the enrollment series for comparability, preserves zero-incident years, and constructs a cleaner state-year panel.



What strengthens the panel

- State and year keys are harmonized across incident, enrollment, and policy sources.
- The enrollment break around 2019 is treated as a comparability issue and rebuilt into a consistent series.
- Zero-incident years remain in the panel, preventing low-incident states from disappearing.

Analytical implications

- The main modeling window is 1987-2025, not the full historical record, because comparability matters more than raw span.
- Incidents are analyzed as risk per student, not raw volume, using enrollment-normalized rates and log offsets.
- The outputs support descriptive, panel-based inference rather than a causal policy claim.

1966-2025
Overall incident history available in the repo outputs.

1987-2025
Core analytic window used for the cleaned state-year panel.

51
States in the panel, including D.C.

Best-covered fields: date, place, and incident count.

Modeling Framework

The Project Uses Multiple Model Types

The modeling work separates state-year incident risk from incident-level severity. Count models explain whether incidents happen more often; severity models explain how harmful incidents are once they occur.

Count-risk models

These models operate on the state-year panel.

Negative Binomial Baseline

Initial count model for the zero-heavy, right-skewed incident_count outcome.

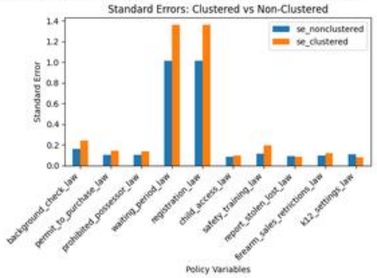
Two-Way Fixed Effects Risk Model

Adds state effects, year effects, and a log enrollment offset so the comparison is incident risk rather than raw volume.

Preferred Final Count Specification

README reports NB2 alpha near zero with unstable convergence, so Poisson with state-clustered errors becomes the final count model.

Clustered inference matters



Severity models

A second Negative Binomial model tracks total victims. An incident-level Gamma GLM models victims per incident; that is where the clearest policy-linked severity signal appears.

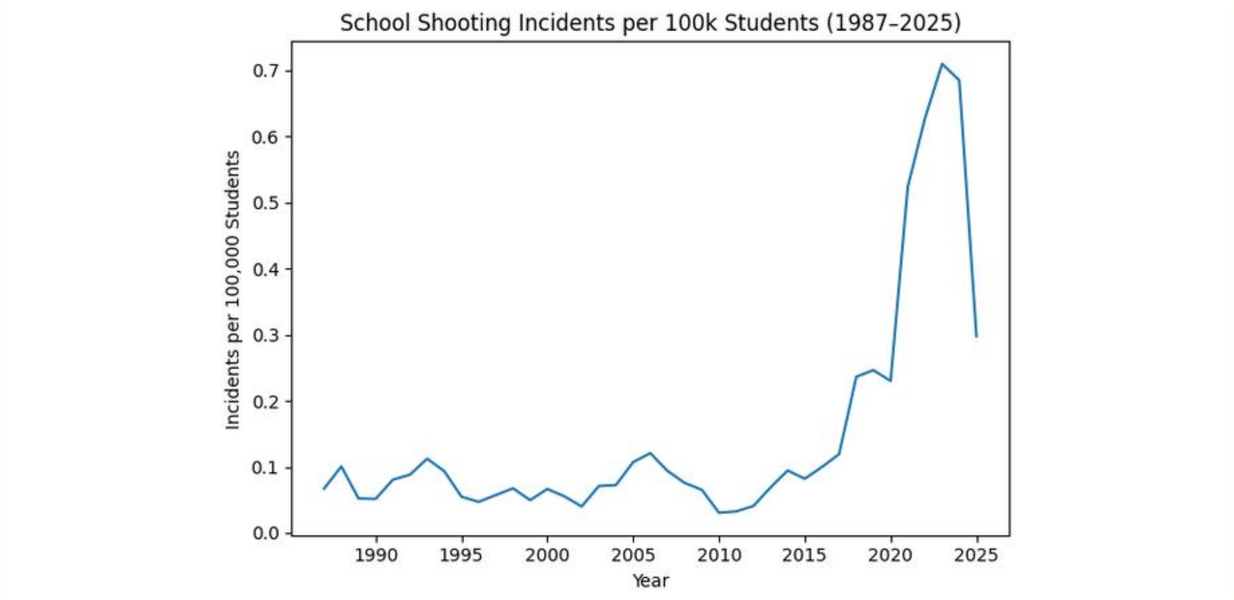
Interpretation: the project compares model families, but the strongest recurring story is still the broader time trend rather than a large set of stable policy effects.

Project Findings

Headline Findings From The Full Project

Across notebooks, scripts, and frozen outputs, the strongest supported pattern is a late-year rise in national incident risk. Most single-policy effects stay smaller and less consistent than the broader time pattern.

National incident risk



Peak observed national rate in the repo summaries: 0.71 incidents per 100,000 students in 2023.

0.71

Peak national rate in the available series, reached in 2023.

2018-2024

Window where the national rise is visually and descriptively strongest.

0.30

Latest 2025 rate reported in the Plotly insight summary.

Main analytical result

- The main incident-count and total-victim models do not show many clean single-policy effects.
- The time pattern is stronger and more stable than most individual policy coefficients.
- The repo supports a stronger trend story than a strong single-policy story.

Recent monitoring lens

- In the repo's burden and volatility view, CA, TX, OH, NY, and IL are the leading recent monitoring states.
- Fast-rising recent trajectories in the Plotly insight summary are FL, NE, and WA.

Severity model note: K-12 settings is the clearest positive victims-per-incident signal in the project.

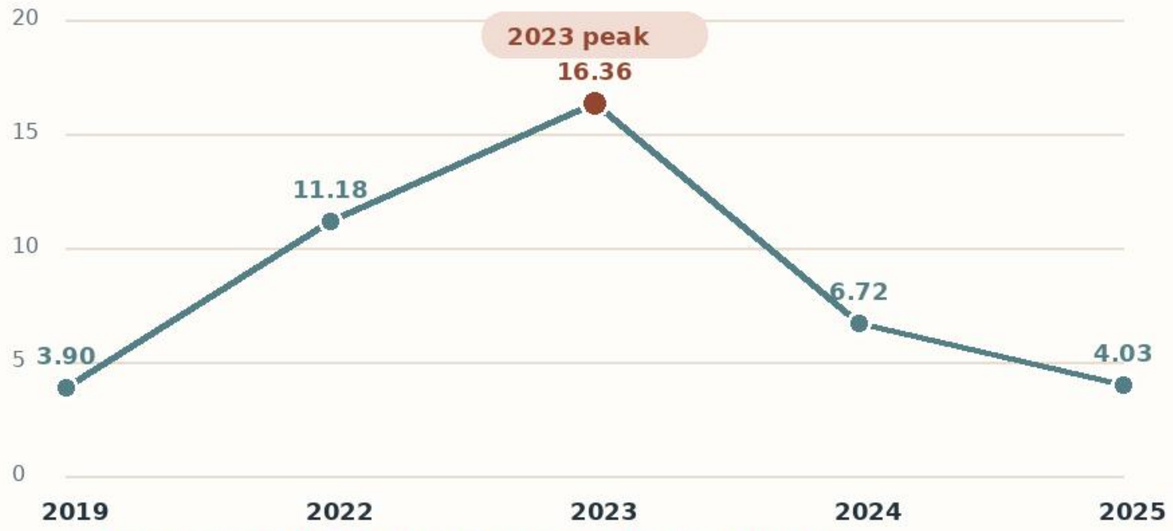
D.C. Findings

D.C. Is A Rate Outlier, But Not A Severity Outlier

The project treats D.C. as a meaningful special case. Rate-based outputs show sharp spikes and a large positive state effect, while the incident-level severity model does not identify D.C. as unusually severe once an incident occurs.

D.C. rate pattern

Selected annual rates pulled directly from the notebooks. The sharpest spike appears in 2023.



Selected annual D.C. incident rates per 100,000 students

Risk-based evidence

- D.C. reaches 16.36 incidents per 100,000 students in 2023, with another large spike in 2022.
- The risk models show a large positive D.C. state effect, around +2.86, with strong statistical significance.
- The SMR notebook also flags D.C. as an extreme outlier with SMR = 10.50.

Severity result

In the incident-level Gamma model for victims per incident, the D.C. coefficient is negative and not statistically significant. That means D.C. stands out more on incident risk than on conditional severity.

Robustness and interpretation

The no-D.C. robustness check leaves the overall policy story broadly intact. The main shift is that report_stolen_lost_law moves to conventional significance, while the larger national time-trend story remains unchanged.

Bottom line: D.C. is a strong rate-based outlier in the project, but it is not the driver of the full-project conclusions and it is not a severity outlier in the Gamma model.